

Homework 4: Qualitative Analysis III — Mixtures & Equations

Paper track. Pairs with Home Lab 4. Do this on your own. Calculator OK.

Part A: Mixture logic (2 pts each)

For each, say **single powder** or **mixture**, and name the component(s):

1. Fizzes strongly, dissolves fully, pH 9.
2. Fizzes weakly, most of it won't dissolve, iodine-negative, gritty residue.
3. Half dissolves; dissolved part is acidic (pH 3); undissolved part turns iodine blue-black.
4. Dissolves fully; two different crystal shapes visible under the lens; neutral; iodine-negative.

Part B: Balancing (1 pt each)

1. $\underline{\quad} \text{H}_2 + \underline{\quad} \text{O}_2 \rightarrow \underline{\quad} \text{H}_2\text{O}$
2. $\underline{\quad} \text{Na} + \underline{\quad} \text{Cl}_2 \rightarrow \underline{\quad} \text{NaCl}$
3. $\underline{\quad} \text{Fe} + \underline{\quad} \text{O}_2 \rightarrow \underline{\quad} \text{Fe}_2\text{O}_3$
4. $\underline{\quad} \text{Mg} + \underline{\quad} \text{HCl} \rightarrow \underline{\quad} \text{MgCl}_2 + \underline{\quad} \text{H}_2$
5. $\underline{\quad} \text{C}_3\text{H}_8 + \underline{\quad} \text{O}_2 \rightarrow \underline{\quad} \text{CO}_2 + \underline{\quad} \text{H}_2\text{O}$
6. $\underline{\quad} \text{NaHCO}_3 + \underline{\quad} \text{H}_3\text{C}_6\text{H}_5\text{O}_7 \rightarrow \underline{\quad} \text{CO}_2 + \underline{\quad} \text{H}_2\text{O} + \underline{\quad} \text{Na}_3\text{C}_6\text{H}_5\text{O}_7$

Part C: Worked example — copy the method (6 pts)

Solved for you: 12 g of baking soda (NaHCO_3 , 84 g/mol) reacts with excess vinegar. How many grams of CO_2 (44 g/mol) are produced? $\text{NaHCO}_3 + \text{CH}_3\text{COOH} \rightarrow \text{NaC}_2\text{H}_3\text{O}_2 + \text{H}_2\text{O} + \text{CO}_2$

- moles $\text{NaHCO}_3 = 12 \div 84 = 0.143 \text{ mol}$
- ratio $\text{NaHCO}_3 : \text{CO}_2$ is 1 : 1 $\rightarrow 0.143 \text{ mol CO}_2$
- mass $\text{CO}_2 = 0.143 \times 44 = \mathbf{6.3 \text{ g}}$

Now you do it: 21 g of baking soda reacts with excess vinegar. How many grams of CO_2 are produced? Show every step.

Part D: Limiting reactant (4 pts)

You mix 0.40 mol of NaHCO_3 (coefficient 3) with 0.10 mol of citric acid (coefficient 1), following $3 \text{ NaHCO}_3 + \text{H}_3\text{C}_6\text{H}_5\text{O}_7 \rightarrow 3 \text{ CO}_2 + 3 \text{ H}_2\text{O} + \text{Na}_3\text{C}_6\text{H}_5\text{O}_7$.

1. Which is the limiting reactant? Show the division.
2. How many moles of CO_2 form?

Answer Key

Part A: 1. Single — baking soda. 2. Mixture — a carbonate (baking soda) + sand (SiO_2 , gritty, insoluble). 3. Mixture — citric acid (soluble, acidic) + corn starch (insoluble, iodine +). 4. Mixture — e.g. table salt + sugar (both soluble, neutral, iodine-negative, different crystal shapes).

Part B: 1) 2,1,2 2) 2,1,2 3) 4,3,2 4) 1,2,1,1 5) 1,5,3,4 6) 3,1,3,3,1

Part C: moles $\text{NaHCO}_3 = 21 \div 84 = 0.25 \text{ mol}$; ratio 1:1 $\rightarrow 0.25 \text{ mol CO}_2$; mass = $0.25 \times 44 = 11 \text{ g CO}_2$.

Part D:

1. NaHCO_3 : $0.40 \div 3 = 0.133$. Citric acid: $0.10 \div 1 = 0.10$. 0.10 is smaller $\rightarrow$ **citric acid is limiting**.
2. CO_2 : citric acid ratio is 3 : 1 $\rightarrow 0.10 \times 3 = 0.30 \text{ mol CO}_2$.